

2003 IRW FINAL REPORT

fuses and read latches, and 504 parallel transfer gates connected to a 504 bit shift register. This macro is one of many supporting product scribe line structures.

Fuse programming data enters from a DQ pad into the shift register through a pair of non-overlapping clocks, CLK and CLK_n. Next fuse data is loaded into each respective fuse latch by phase LOADP. The programming voltage FSOURCE is then applied to program selected fuses. The read out operation is similar with fuse data sensed by the fuse latches and subsequently transferred to the output serial shift register.

The BPM allows for a large quantity of programming and reliability data to be collected by an area-and pin-efficient method. This BPM was incorporated into the scribe of a 2.5V 0.14μm CMOS 256M SDRAM technology, and contains the edge seal and crack stop layers of the standard SDRAM product chip. The BPM is packaged into 400 mil wide, 66-pin TSOPs. On a given chip the BPM contained E-Fuse design A or B, but not a mixture of A and B.

PROGRAMMING OPTIMIZATION

Figures 5a to 5c indicate some of the initial experiments used to determine the fuse programming conditions of E-Fuse B. Figure 5a shows the experimental results of varying the program voltage condition between 2.4V and 4.8V, maintaining a constant programming

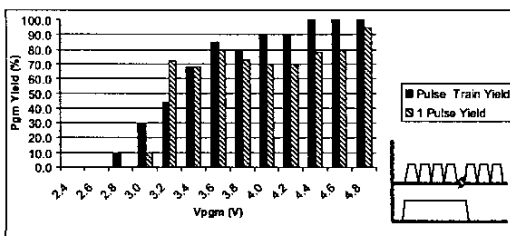


Figure 5a: Time zero E-Fuse B determination of programming voltage and pulse train effectiveness for a total program time of 250μS.

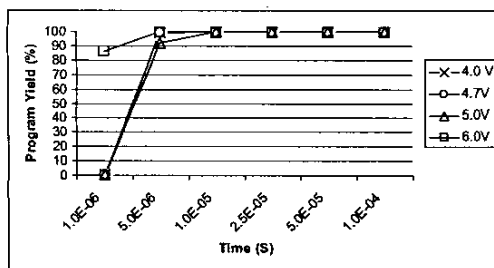


Figure 5b: Time zero E-Fuse B determination of programming time for short pulse durations

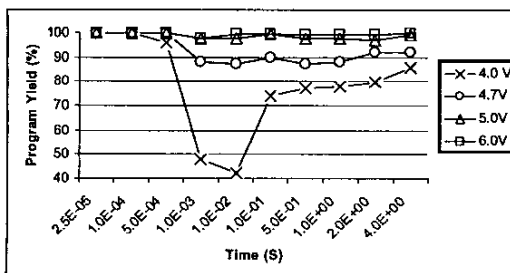


Figure 5c: Time zero E-Fuse B determination of programming time for long pulse durations.

time of 250μS. A single pulse of 250μS was contrasted against a pulse programming train of 25 pulses of 10μS each. We experimentally observed a given pulse train is more effective than a fixed pulse in developing a high yield E-Fuse.

Figures 5b and 5c indicate a typical programming resultant study varying (in this case) the programming fuse voltage. The dependant parameter along the horizontal axis is programming time. Fuse programming current was maintained at 5mA. The goal of this experiment is to sweep the range between "cold and "hot" fuse programming. Cold programming equates to short durations of ~1mS to ~10mS pulse times. Hot side programming is equated to programming durations in excess of ~1mS. We found the cold end of the spectrum at a V_{dd} of 6V was advantageous. Although Figure 5b indicates successful programming at this condition the programming transistor is not reliable. To implement this condition reliably in this technology one would have to externally supply the E-Fuse programming voltage. This was not desirable from the perspective of field repair, and we chose a lower internal voltage and a longer pulse duration and train for final programming.

Ultimately the time zero programming yield is used to determine an E-Fuse schmoo operational window. Additional studies where programming voltage, fuse ambient temperature, and various alternate pulse trains as the respective independent variable were evaluated, culminating in an optimum program environment and electrical stimulus. In addition to the problems associated with high voltage or cold side programming, low voltage or hot side programming generally correlates to large width programming transistors supplying adequate current under long duration. These attributes are also undesirable for E-Fuse implementations.

In this technology a dual oxide thickness was employed where the programming transistor was capable of 10 year reliable operation at 5.0 volts. A matrix of ~1000 E-Fuses per design type across the process parameters is used to validate the programming conditions for each experiment.

Figure 6 indicates a typical pre and post stress design window for a type B E-Fuse. The program resistance is indicative of an open circuit. In this figure the intrinsic resistance of an unprogrammed fuse is shown (~350Ω) and the combined unprogrammed resistance

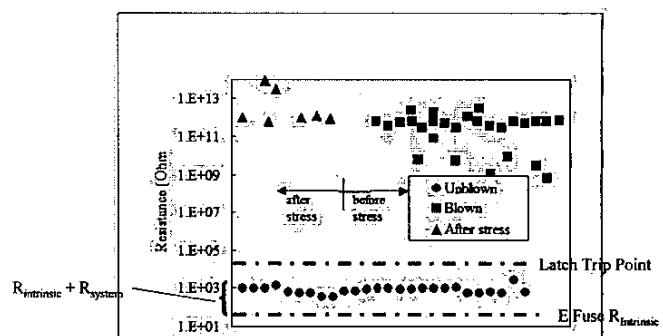


Figure 6: Pre and Post Stress E-Fuse Resistance

of all the elements in the path are also shown (~10kΩ). The choice of a latch trip point must be able to distinguish between these unprogrammed and programmed states.

This electrical result, i.e. the time zero composite resistance of ~10kΩ and the infinite resistance of a programmed state can be thought of as a digital resistor, having no tail elements when programmed. This observation is unlike any E-Fuse to date, where the program result approaches that of a laser fuse.

RELIABILITY EVALUATION

Hardware was generated across several wafer lots having intentional 3 variations in the critical parameters of Lpoly, VT, and gate oxide thickness in order to qualify both the E-Fuse programming and latching circuitry with adequate guardband against process variation. Time zero programming conditions were established by investigating the programming yield using a representative statistical sample from the process splits. ~150,000 fuses were pre-programmed and stressed from each sample, and ~150,000 fuses were left un-programmed. All parts were preconditioned per JEDEC standard requirements [5] prior to stress, as shown in Table 1. Table 2 lists the stress conditions that were applied to samples having the odd fuses programmed, or the even fuses programmed, or all fuses programmed. "Even" and "odd" were indicative of the fuse position in the physical bit order within the BPM. In addition, odd fuses had Metal 1 running overhead, while even fuses did not.

The "all programmed" pattern was required for HAST in order to maintain a constant bias across the fuses, after setting the support circuits appropriately. The "even" and "odd" patterns were required to evaluate the risk of collateral damage to unprogrammed fuses, from the fusing process. The alternate read-write stress was similar to a device HTOL stress [6], with the fuses being read at ~ 50% temporal duty factor. The temperature cycling stress [7] was required to verify mechanical stability. HAST [8] was required to evaluate the corrosion risk on programmed fuses.

The fuse yield at time zero, after preconditioning, and after stress is shown in Table 3. E-Fuse B had poor results at time zero and after preconditioning, and the stresses were not completed. E-Fuse B had perfect results at every checkpoint.

E-Fuse lifetime estimates based on typical IBM DRAM and high performance SRAM applications are shown in Table 4. The SRAM application is more stringent as fuses are read every "x" cycles, as opposed to only at powerup for a DRAM. Lifetime estimates are obtained by multiplying the zero-fail stress duration times an acceleration factor. The acceleration factors for the biased stresses (RW2, HAST) are determined solely by electrical duty factor considerations. Since no stress fails are observed, acceleration factors for voltage and temperature are not able to be determined. However, those values would likely be greater than 1 since the fuse operating voltage and device temperature (T_j) in the application would be lower than the stress condition. Both HAST and RW2 lifetimes are thus pessimistically determined solely based upon stress hours as compared to use hours. For example, the HAST SRAM stressing shown in Table 4 (130°C/2.85V/192 hr, 100% duty) equating to 1.5E6 device hours is determined by:

(1) $Lifetime = fuseduty \times stress\ time$
where:

$$fuseduty(\# \text{ sec} / \text{ hr}) = \frac{(3600 / T_{cy})}{Clk_{dty}}$$

$$T_{cy} = 4\text{ns/cycle}, Clk_{dty} = 1000 \text{ cycles}, stress\ time = 192\text{hr}.$$

Table 1: Sample preconditioning prior to stress

Step	Conditions
Bake	125°C/24 hr
Shippability Temp. Cycling	-40/60°C, 10 cycles at 1 cph
Bake	125°C/24 hr
Soak	30°C/60%RH, 8 days
Reflow	3X convection, 235°C peak

Table 2: Stress Description

Stress	Conditions	SS (Lots)	Fuse Pattern	Readouts
Alt.Read Write RW2	140°C, 2.85V, est. 50% duty	443 (5)	Odd	168, 500 hr
Temperature Cycling	-55/125°C air-air	84 (1)	Even	140, 500 cy
HAST	130°C, 85%RH, 1.9V	20 (1)	Even then Odd	96, 192 hr
HAST	130°C, 85%RH, 2.85V	40 (1)	Even then Odd	96, 192 hr

Table 3 Yield

E-Fuse Type	T-0	Post Precon	Stress
A	80%	99.97%	Not Completed
B	100%	100%	100%

Table 4 E-Fuse Lifetime Estimates

Stress	Conditions	Equivalent Product Lifetime		Rationale
		SRAM-a	DRAM-b	
HAST	130°C/2.85V/192 hr, 100% duty	1.5E6 device hours	2.4E9 years	Electrical Duty-c
RW2	140°C/2.85V/500 hr, 50% duty	2E6 device hours	3.1E9 years	Electrical Duty-c
DTC	-55/125°C/500 cycles	43 years	43 years	Coffin-Manson model -d

a) Assumes 500 ps fuse query per 1000 clock cycles at 4 ns cycle

b) Assumes 400 ns fuse query at powerup, and 2 powerups per day

c) Temperature and voltage acceleration not included

d) Assumes minimum exponent of 5.5 for e.g. thin film cracking, $\Delta T(\text{field}) = 85^\circ\text{C}$ e.g. $T_j = 105^\circ\text{C}$ and $T_{off} = 20^\circ\text{C}$, and 2 on/off per day

In this case temperature and voltage acceleration are ignored.

The acceleration factor for temperature cycling is determined by postulating a "thin-film cracking" mechanism and using a published exponent [9] for that mechanism in the simple Coffin-Manson equation [10]. This pessimistic reliability projection is the limiting case in Table 4. It is estimated as follows:

$$(2) \text{ The acceleration factor } AF = \frac{\Delta T_{stress}^\alpha}{\Delta T_{field}}$$

Where $\Delta T_{stress} = 180$, $\Delta T_{field} = 85$, and $\alpha = 5.5$

Combining (1) with the actual stress of 500 cycles, and assuming 2 user powerups per day results in a lifetime of 43 years as indicated in Table 4.

PHYSICAL ANALYSIS AND MECHANISM

The method of programming either E-Fuse A or E-Fuse B occurs when the WSi₃ layer reaches approximately 2200°C and migrates into the fuse neck, leaving behind a highly resistive polysilicon denuded

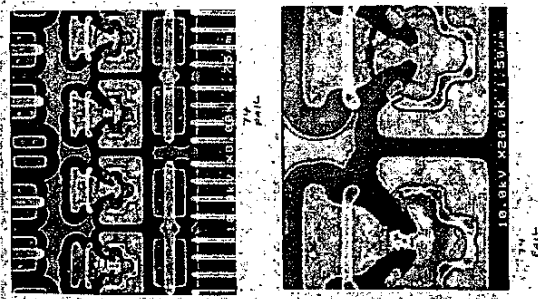
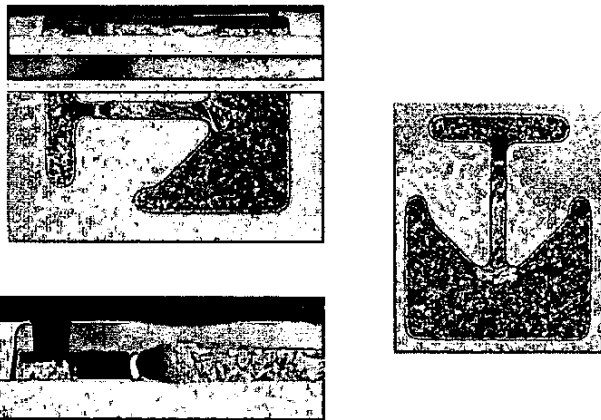
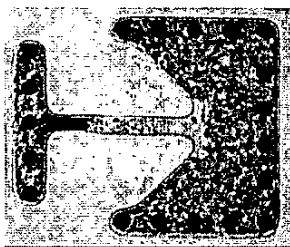


Figure 7: E-Fuse A — Physical Failure Analysis

Figure 8: E-Fuse B post programming result. Complete programming for 250 μ S, 4.7V. Three fuses shown.Figure 9: E-Fuse B post programming result. Incomplete programming for 25 μ S, 4.7V

zone. The denuded polysilicon becomes more resistive by self heating and grain recrystallization, resulting in a stable, high resistance programmed state. Kalnistky [4] described an E-Fuse similar to the type A design, in which the underlying polysilicon was intrinsic. The use of intrinsic polysilicon requires a prohibitively large transistor aspect ratio in relation to the E-Fuse to maintain adequate current flow during the programming operation. To

enable fuse self heating, we place a low IR drop sink and a source of WSi_2 . This lead to the subsequent design methodologies of fuse A and

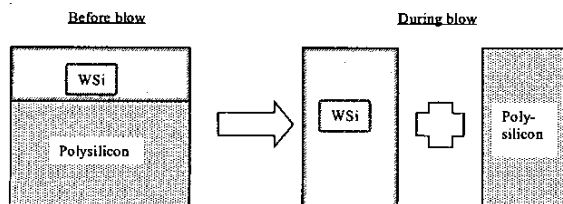


Figure 10: E-Fuse Programming Model

B. It is the intent to electrically open the remaining polysilicon by self heating and grain re-crystallization, thus providing a high resistive programmed state that is digital in nature, and does not rely on the latches internal ability to distinguish a minimum 30KW programmed state. This relieves the design from the perspective of analyzing an analog element, and affords a much smaller unit cell. If the resultant program resistance contained a tail, one might consider differentially sensing the programmed resistance effectively increasing the E-Fuse cell area.

Physical failure analysis of E-Fuse A showed the neck design allowed thermal spread into the adjoining anode and cathode regions, resulting in incomplete programming (Figure 7).

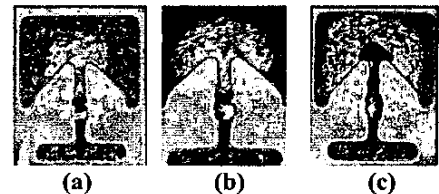


Figure 11: E-Fuse B Overprogramming result. 4.7V, 5mA (a) T=1mS, (b) T=2S, (c) T=4S

The design changes in E-Fuse B resolved the heat distribution problems. Figure 8 shows a plan view TEM and a cross section of a type B fuse that was programmed for 250 μ S at 4.7 V, and as indicated in Figure 5b this is a high yielding fuse corner. Figure 5a and Figure 9 displays the sensitivity to programming yield with respect to short times. The Figure 5a fuse has a programming yield of 100% for ea. 1000 fuses. The Figure 9 fuse with a 25 μ S programming pulse at 4.7 V did not program. The Figure 9 fuse was part of a very large sample used to investigate the yield as a function of programming conditions.

An attempt was made to explore a physical programming model where the silicide layer is first heated by the programming device current that flows predominately within the silicide (Fig. 10). As the silicide heats and reaches its liquid state it begins to migrate and collect in the fuse neck, where now the current flow is out of the silicide, down into the N++ polysilicon, and back up to a new silicide composition, possibly W_2Si or W_3Si_2 . In this model programming takes place after the driving force is removed from the fuse and the fuse cools. This is explored by the time slide of fuse programming shown in Figures 5c and 11, where the programming time was varied from 1mS to 4S, well beyond the characterized nominal value of 250 μ S. The physical analysis data of figure 11 indicates as the program time is increased, current continues to flow through the fuse, and the WSi_2 continues to migrate, damaging the fuse and its superstructure forming unwanted failing filaments around the neck. This evidence indicates the proposed physical model has validity.

Finally, when choosing a programming voltage, current, and duration one must take into account the statistical parameters (both systematic and independent variables) that will affect the programming condition and either choose a single value that satisfies all conditions, or tune the thermal stimulus to the in line process measurements. For this fuse the key variables that affected the choice of thermal stimulus are: Lpoly variation, and the programming device oxide thickness variation.

CONCLUSION

In summary, E-Fuses can be designed to operate reliably. The fuse geometry must be optimized in order to function within the required fuse programming conditions. The parameters that generally affect device and circuit performance also affect the ability for a fuse to program, and also affect the necessary and sufficient stimulus that must be applied either from an internally generated circuit, or from an external source. Characterizing the process for it's variability is an important step in the miniaturization of the fuse element, the programming device, and the output latch. An over aggressive design

may either fail for reliability or require extensive on die corrections for lot to lot tuning, and is generally thought to be a poor choice in a final E-Fuse design.

An electrically programmable fuse with on-chip support circuitry was designed, fabricated, programmed, and evaluated. The first design, E-Fuse A, has poor programming yield and an inadequate preconditioning survival rate. Physical failure analysis determined the cause as deleterious heat distribution during programming. The second design, E-Fuse B [11], corrected the problem of E-Fuse A by redesign of the fuse neck. E-Fuse B subsequently survived a rigorous reliability evaluation. This design and process has since been successfully integrated into IBM semiconductor products. The development and qualification methodology used for the IBM E-Fuse can be applied to other fuse types and designs.

REFERENCES

- [1] G. Shirley, S Defect Reliability Statistics with Redundancy 2002 IRPS Tutorial, topic
- [2] E.G. Colgan, J.P. Gambino, Q. Z. Hong: Formation and stability on silicides on polycrystalline silicon, Materials Science and Engineering, R16, No. 2, February 1996
- [3] J.P. Gambino, E.G. Colgan, A. G. Domenicucci, B. Cunningham: The Thermal Stability of CoSi₂ on Polycrystalline Silicon: The Effect of Silicon Grain Size and Metal Thickness
- [4] A. Kalnitsky, I Saadat, A Bergemont, P Francis: CoSi₂ integrated fuses on polysilicon for low voltage 0.18 μ m CMOS applications, 1999 IEDM
- [5] JESD-22 A113, Preconditioning of Nonhermetic Surface Mount Devices Prior to Reliability Testing
- [6] JESD-22 A108, Temperature, Bias, and Operating Life
- [7] JESD-22 A104, Temperature Cycling
- [8] JESD-22 A110, Highly Accelerated Temperature and Humidity Stress Test (HAST)
- [9] EIA Bulletin SSB-1
- [10] JEP-122, Failure Mechanisms and Models for Silicon Semiconductor Devices
- [11] U.S. patent pending.

QUESTIONS AND ANSWERS

Q1. Does your proposed model account for the melting silicide?

Answer: The model follows the physical change in the Polysilicon, Tungsten Silicide Stack. During programming the majority of the current flow is through the Tungsten Silicide layer that migrates along the Polysilicon line in the direction of electron current flow. The Silicide layer does not melt, but it heats the underlying polysilicon layer.

Q2. What is actually melting: Tungsten Silicide or Polysilicon?

Answer: The polysilicon is the layer that actually melts and separates, taking with it the tungsten silicide. As seen in the TEM micrographs, a region about the separation indicates the Polysilicon has recrystallized, and has undergone localized heating.